Solar Inverter



Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Inverter (String / Hybrid); Solar photovoltaic inverter

What it is used for

A solar inverter converts DC electricity from solar panels or batteries into usable AC electricity for compatible loads or the grid. Correct installation checks, operating limits, and monitoring help maintain reliable performance and reduce electrical hazards.

Main parts

- DC input terminals receive power from the solar array or battery where supported.
- Power conversion circuit converts DC electricity into AC electricity.
- AC output terminals connect the inverter to approved loads or the grid.
- MPPT controller optimises solar panel operating conditions on equipped models.
- Display or monitoring interface shows operating values and fault codes.
- Cooling system removes heat from internal components.
- Protection circuits respond to electrical faults and abnormal conditions.
- Disconnect switches and protective devices isolate circuits where provided.

Before you start

1. Only qualified and authorised personnel should install or service the inverter.
2. Read the manual and confirm compatibility with the solar array, battery, and AC system.
3. Inspect the enclosure, cables, connectors, isolators, and visible wiring for damage.
4. Check that voltage, current, polarity, and power ratings are within the manufacturer's limits.
5. Confirm the mounting location is dry, ventilated, and suitable for the equipment rating.
6. Verify earthing, protective devices, and required isolation arrangements according to the approved installation design.

How to use

1. Keep the inverter isolated from all energy sources before making or changing connections.
2. Verify DC and AC cable identification, polarity, terminal ratings, and protective arrangements.
3. Connect the solar array, battery if supported, and AC circuits according to the approved wiring diagram.
4. Secure the terminals to the specified torque using the correct tools.
5. Check that covers, cable glands, earthing, and protective devices are correctly installed.
6. Restore power in the sequence specified by the manufacturer and approved installation procedure.
7. Check the display for normal startup, operating status, and any fault codes.
8. Confirm that measured voltage, current, and operating conditions are within the specified limits.

9. Monitor generation and alarms through the display or approved monitoring system.
10. If a fault occurs, follow the shutdown procedure and arrange qualified inspection before restarting.

⚠ Safety

- Solar DC conductors may remain live in daylight even when the inverter is switched off.
- Never open the enclosure or touch internal terminals while energised.
- Use appropriately rated PPE and test instruments for the electrical system.
- Never disconnect DC connectors under load unless the equipment procedure explicitly permits it.
- Do not bypass earthing, isolation, overcurrent protection, or other safety devices.
- Follow the manufacturer's shutdown sequence and wait the specified time before servicing.

Care & storage

- Keep ventilation openings clear and remove external dust using the approved method.
- Inspect visible cables, connectors, glands, and enclosure seals at recommended intervals.
- Review fault logs and generation data for abnormal changes.
- Arrange authorised servicing for repeated faults, overheating, unusual noise, or damaged components.

Common mistakes to avoid

- Connecting DC cables with incorrect polarity or unsuitable ratings.
- Ignoring earthing, isolation, or protective device requirements.
- Blocking ventilation openings and allowing excessive heat buildup.
- Opening the inverter before all energy sources are safely isolated.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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